Math 49 Curriculum

Precalculus with Limits | Middlesex School Mathematics | Textbook: OpenStax Precalculus 2e (Jay Abramson, 2021); OpenStax Calculus Volume 1, Chapter 2, "Limits" (Gilbert Strang and Edwin Herman)

COURSE AT A GLANCE

Review Content

REVIEW

1.2 Domain and Range; 1.5 Transformation of Functions; 4.1 Exponential Functions; 4.2 Graphs of Exponential Functions; 4.7 Exponential and Logarithmic Models; 4.3 Logarithmic Functions; 4.4 Graphs of Logarithmic Functions; 4.5 Logarithmic Properties

UNIT LEARNING OBJECTIVES

I can determine the domain and range of a function from a formula, graph, or table.

I can describe and apply transformations of a parent function.

I can analyze exponential functions using their equations, graphs, and contexts.

I can analyze logarithmic functions using their equations, graphs, and contexts.

I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

Function Operations, Composition, and Inverses

REQUIRED

1.4 Composition of Functions; 1.7 Inverse Functions

UNIT LEARNING OBJECTIVES

I can determine whether a function is even, odd, or neither.

I can add, subtract, multiply, and divide functions and state the resulting domain.

I can evaluate a composition from formulas, graphs, or tables.

I can write a formula for the composition of two functions and determine its domain.

I can decompose a function into a composition of simpler functions.

I can determine whether a function has an inverse on a given domain.

I can find and verify an inverse function algebraically.

I can interpret the relationship between a function and its inverse from graphs or tables.

Power Functions

REQUIRED

3.3 Power Functions and Polynomial Functions; 3.9 Modeling Using Variation; 3.8 Inverses and Radical Functions

UNIT LEARNING OBJECTIVES

I can identify a power function and interpret the parameters in y = kx^p.

I can construct a power function from two data points or a proportional relationship.

I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

I can recognize and model direct, inverse, and other power-variation relationships.

I can solve equations involving integer or rational powers.

I can build or select a power-function model for a context and interpret its parameters.

I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.

Polynomial Functions

REQUIRED

3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions

UNIT LEARNING OBJECTIVES

I can predict the end behavior of a polynomial from its degree and leading coefficient.

I can determine polynomial zeros and their multiplicities.

I can construct a polynomial function from specified zeros or graphical features.

I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.

I can determine the minimum possible degree of a polynomial function from the zeros, turning points, inflection points, and end behavior of its graph.

I can build or select a polynomial model for a context and interpret its features.

Rational Functions and Limits

REQUIRED

3.7 Rational Functions; 12.1 Finding Limits: Numerical and Graphical Approaches; 2.1 A Preview of Calculus; 2.2 The Limit of a Function; 2.3 The Limit Laws; 3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions; 4.4 Graphs of Logarithmic Functions; Chapter 2 Limits; 3.9 Modeling Using Variation

UNIT LEARNING OBJECTIVES

I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.

I can sketch and analyze a rational function using its algebraic and asymptotic features.

I can interpret limit notation as a statement about function behavior near an input.

I can estimate a limit from a graph or table.

I can evaluate a limit algebraically using direct substitution or simplification.

I can interpret an infinite limit as vertical-asymptotic behavior.

I can use limit notation to describe behavior at infinity.

I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions, and combinations thereof.

I can build or select a rational or rational-adjacent model for a context and interpret its long-run behavior.

Sequences and Series

REQUIRED

11.1 Sequences and Their Notations; 11.2 Arithmetic Sequences; 11.3 Geometric Sequences; 11.4 Series and Their Notations

UNIT LEARNING OBJECTIVES

I can evaluate and graph a sequence defined explicitly or recursively.

I can write explicit and recursive formulas for arithmetic sequences.

I can write explicit and recursive formulas for geometric sequences.

I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.

I can create and interpret a sequence model for discrete change.

I can interpret and use summation notation.

I can calculate a finite arithmetic series.

I can calculate a finite geometric series.

I can determine whether an infinite geometric series converges and find its sum when it does.

I can create and interpret a series model for accumulated change.

Additional Limit Concepts

EXTENSION

12.1 Finding Limits: Numerical and Graphical Approaches; 2.1 A Preview of Calculus; 2.2 The Limit of a Function; 2.4 Continuity

UNIT LEARNING OBJECTIVES

I can determine one-sided limits and decide whether a two-sided limit exists.

I can distinguish a function value from a limit and determine continuity at a point.

Parametric Equations

EXTENSION

8.7 Parametric Equations: Graphs; 8.6 Parametric Equations

UNIT LEARNING OBJECTIVES

I can graph a plane curve defined by parametric equations with its orientation.

I can eliminate a parameter to obtain a rectangular equation.

I can create or interpret parametric equations for motion in the plane.

Conic Sections

EXTENSION

10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates; 10.3 The Parabola

UNIT LEARNING OBJECTIVES

I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.

I can use completing the square to rewrite a general-form conic equation in standard form.

I can write, interpret, and graph implicit and parametric equations of a circle.

I can write, interpret, and graph implicit and parametric equations of an ellipse.

I can write, interpret, and graph implicit and parametric equations of a hyperbola.

I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.

Conic Sections (Continued)

EXTENSION

UNIT LEARNING OBJECTIVES - CONTINUED

I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.

I can explain a parabola as the set of points equidistant from a focus and directrix.

IN-DEPTH CURRICULUM

SKILL PROGRESSION I Introduce R Reinforce D Deepen A Apply

Review Content

REVIEW

Textbook coverage: 1.2 Domain and Range; 1.5 Transformation of Functions; 4.1 Exponential Functions; 4.2 Graphs of Exponential Functions; 4.7 Exponential and Logarithmic Models; 4.3 Logarithmic Functions; 4.4 Graphs of Logarithmic Functions; 4.5 Logarithmic Properties

M49-FND-001

I can determine the domain and range of a function from a formula, graph, or table.

Textbook: 1.2 Domain and Range

SUPPORTING SKILLS

- D Determine domain restrictions from denominators, even roots, and logarithms.
  SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- R Read domain and range from a graph.
  SK-FUNC-DOMAIN-RANGE-GRAPH · first introduced in Math 12
- R Read domain and range from a table.
  SK-FUNC-DOMAIN-RANGE-TABLE · first introduced in Math 12

M49-FND-002

I can describe and apply transformations of a parent function.

Textbook: 1.5 Transformation of Functions

SUPPORTING SKILLS

- R Interpret transformations applied to function inputs.
  SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31
- R Distinguish stretches from compressions and horizontal effects from vertical effects.
  SK-FUNC-TRANSFORM-SCALE · first introduced in Math 31
- R Interpret transformations applied to function outputs.
  SK-FUNC-TRANSFORM-VERTICAL · first introduced in Math 31

M49-FND-004

I can analyze exponential functions using their equations, graphs, and contexts.

Textbook: 4.1 Exponential Functions; 4.2 Graphs of Exponential Functions; 4.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

- R Identify horizontal asymptotic behavior of an exponential graph.
  SK-EXP-ASYMPTOTE · first introduced in Math 31
- R Identify initial amount, rate, and time interval in context.
  SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- A Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M49-FND-005

I can analyze logarithmic functions using their equations, graphs, and contexts.

Textbook: 4.3 Logarithmic Functions; 4.4 Graphs of Logarithmic Functions; 4.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

- R Identify vertical asymptotic behavior of a logarithmic graph.
  SK-LOG-ASYMPTOTE · first introduced in Math 31
- R Interpret a logarithm as an exponent.
  SK-LOG-DEFINITION · first introduced in Math 31
- A Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M49-FND-006

I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

Textbook: 4.3 Logarithmic Functions; 4.5 Logarithmic Properties; 4.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

- R Rewrite exponential expressions using a common base when possible.
  SK-EXP-REWRITE-BASE · first introduced in Math 31
- A Reject solutions that make a logarithm argument nonpositive.
  SK-LOG-DOMAIN-CHECK · first introduced in Math 31
- A Apply the logarithm power property in either direction.
  SK-LOG-POWER · first introduced in Math 31
- A Apply the logarithm product property in either direction.
  SK-LOG-PRODUCT · first introduced in Math 31
- A Apply the logarithm quotient property in either direction.
  SK-LOG-QUOTIENT · first introduced in Math 31
- R Isolate an exponential expression and apply a logarithm.
  SK-LOG-SOLVE-EXP · first introduced in Math 31
- R Combine or isolate logarithms before converting to exponential form.
  SK-LOG-SOLVE-LOG · first introduced in Math 31

Function Operations, Composition, and Inverses REQUIRED

Textbook coverage: 1.4 Composition of Functions; 1.7 Inverse Functions

M49-FND-003

I can determine whether a function is even, odd, or neither.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- D Test a formula or graph for even or odd symmetry.
  SK-FUNC-SYMMETRY · first introduced in Math 31

M49-OPS-001

I can add, subtract, multiply, and divide functions and state the resulting domain.

Textbook: 1.4 Composition of Functions

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I Determine the domain of a sum, difference, product, quotient, or composition.
  SK-FUNC-DOMAIN-COMBINE
- I Add, subtract, multiply, and divide function outputs.
  SK-FUNC-OPERATIONS

M49-OPS-002

I can evaluate a composition from formulas, graphs, or tables.

Textbook: 1.4 Composition of Functions

SUPPORTING SKILLS

- D Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- D Follow intermediate values to compose functions from tables.
  SK-FUNC-COMPOSE-TABLE · first introduced in Math 31
- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- R Read and write notation for function composition and inverse functions.
  SK-FUNC-STRUCTURE-NOTATION · first introduced in Math 31

M49-OPS-003

I can write a formula for the composition of two functions and determine its domain.

Textbook: 1.4 Composition of Functions

SUPPORTING SKILLS

- D Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- D Determine the domain of a sum, difference, product, quotient, or composition.
  SK-FUNC-DOMAIN-COMBINE · first introduced in Math 39

M49-OPS-004

I can decompose a function into a composition of simpler functions.

Textbook: 1.4 Composition of Functions

SUPPORTING SKILLS

- A Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- I Identify simpler inner and outer functions in a composition.
  SK-FUNC-DECOMPOSE

M49-OPS-005

I can determine whether a function has an inverse on a given domain.

Textbook: 1.7 Inverse Functions

SUPPORTING SKILLS

- D Apply the horizontal line test for invertibility.
  SK-FUNC-INVERSE-HLT · first introduced in Math 31
- R Test whether a function is one-to-one on a domain.
  SK-FUNC-ONE-TO-ONE · first introduced in Math 31

M49-OPS-006

I can find and verify an inverse function algebraically.

Textbook: 1.7 Inverse Functions

SUPPORTING SKILLS

- A Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- D Exchange variables and solve to obtain an inverse formula.
  SK-FUNC-INVERSE-ALGEBRA · first introduced in Math 31
- D Verify inverse functions using composition.
  SK-FUNC-INVERSE-VERIFY · first introduced in Math 31

M49-OPS-007

I can interpret the relationship between a function and its inverse from graphs or tables.

Textbook: 1.7 Inverse Functions

SUPPORTING SKILLS

- D Exchange domain and range between a function and its inverse.
  SK-FUNC-INVERSE-DOMAIN-RANGE · first introduced in Math 31
- D Exchange input and output values when interpreting an inverse.
  SK-FUNC-INVERSE-SWAP · first introduced in Math 31
- D Reverse input and output units when interpreting an inverse.
  SK-FUNC-INVERSE-UNITS · first introduced in Math 31

Power Functions

REQUIRED

Textbook coverage: 3.3 Power Functions and Polynomial Functions; 3.9 Modeling Using Variation; 3.8 Inverses and Radical Functions

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M49-PWR-001

I can identify a power function and interpret the parameters in y = kx^p.

Textbook: 3.3 Power Functions and Polynomial Functions

SUPPORTING SKILLS

- D Interpret the coefficient and exponent in a power function.
  SK-POWER-PARAMETER-INTERPRET · first introduced in Math 39
- R Recall and correctly use power-function terms including coefficient, exponent, direct variation, and inverse variation.
  SK-POWER-VOCABULARY · first introduced in Math 39
- D Represent and interpret a power-variation relationship.
  SK-VAR-POWER · first introduced in Math 39

M49-PWR-002

I can construct a power function from two data points or a proportional relationship.

Textbook: 3.9 Modeling Using Variation

SUPPORTING SKILLS

- A Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- D Determine a power function from two points or a proportional relationship.
  SK-POWER-CONSTRUCT · first introduced in Math 39

M49-PWR-003

I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

Textbook: 3.3 Power Functions and Polynomial Functions

SUPPORTING SKILLS

- D Determine power-function behavior as the input approaches zero or positive or negative infinity.
  SK-POWER-END-BEHAVIOR · first introduced in Math 39
- D Relate the coefficient and exponent of a power function to its graph shape.
  SK-POWER-SHAPE · first introduced in Math 39
- A Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M49-PWR-004

I can recognize and model direct, inverse, and other power-variation relationships.

Textbook: 3.9 Modeling Using Variation

SUPPORTING SKILLS

- D Represent and interpret a direct-variation relationship.
  SK-VAR-DIRECT · first introduced in Math 39
- D Represent and interpret an inverse-variation relationship.
  SK-VAR-INVERSE · first introduced in Math 39
- D Represent and interpret a power-variation relationship.
  SK-VAR-POWER · first introduced in Math 39

M49-PWR-005

I can solve equations involving integer or rational powers.

Textbook: 3.8 Inverses and Radical Functions

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D Interpret numerator and denominator roles in a rational exponent.
  SK-EXP-RATIONAL · first introduced in Math 21

M49-PWR-007

I can build or select a power-function model for a context and interpret its parameters.

Textbook: 3.3 Power Functions and Polynomial Functions; 3.9 Modeling Using Variation

SUPPORTING SKILLS

- I Select a function family whose behavior fits a table, graph, or context.
  SK-MODEL-FAMILY-SELECT
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A Determine a power function from two points or a proportional relationship.
  SK-POWER-CONSTRUCT · first introduced in Math 39

M49-PWR-008

I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.

Textbook: 3.3 Power Functions and Polynomial Functions; 3.8 Inverses and Radical Functions

SUPPORTING SKILLS

- A Interpret numerator and denominator roles in a rational exponent.
  SK-EXP-RATIONAL · first introduced in Math 21
- A Require an even-root radicand to be nonnegative when determining domain.
  SK-FUNC-DOMAIN-EVEN-ROOT · first introduced in Math 21
- I Analyze the domain, graph shape, and behavior near zero of a power function with a fractional exponent.
  SK-POWER-FRACTIONAL-ANALYZE

Polynomial Functions REQUIRED

Textbook coverage: 3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions

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M49-ALG-002

I can predict the end behavior of a polynomial from its degree and leading coefficient.

Textbook: 3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions

SUPPORTING SKILLS

- I Infer polynomial end behavior from degree and leading coefficient.
  SK-POLY-END-BEHAVIOR
- R Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M49-ALG-003

I can determine polynomial zeros and their multiplicities.

Textbook: 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

- I Relate zero multiplicity to local graph behavior.
  SK-POLY-MULTIPLICITY
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12
- D Determine real and complex zeros using an appropriate method.
  SK-POLY-ZEROS · first introduced in Math 21

M49-ALG-004

I can construct a polynomial function from specified zeros or graphical features.

Textbook: 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

- D Build a polynomial from zeros, multiplicities, and a point.
  SK-POLY-CONSTRUCT · first introduced in Math 31
- A Relate zero multiplicity to local graph behavior.
  SK-POLY-MULTIPLICITY · first introduced in Math 39
- A Determine real and complex zeros using an appropriate method.
  SK-POLY-ZEROS · first introduced in Math 21

M49-ALG-005

I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.

Textbook: 3.4 Graphs of Polynomial Functions; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

- D Infer polynomial end behavior from degree and leading coefficient.
  SK-POLY-END-BEHAVIOR · first introduced in Math 39
- D Relate zero multiplicity to local graph behavior.
  SK-POLY-MULTIPLICITY · first introduced in Math 39
- A Determine real and complex zeros using an appropriate method.
  SK-POLY-ZEROS · first introduced in Math 21

M49-POL-002

I can determine the minimum possible degree of a polynomial function from the zeros, turning points, inflection points, and end behavior of its graph.

Textbook: 3.4 Graphs of Polynomial Functions

SUPPORTING SKILLS

- I — Use inflection points to establish a minimum possible polynomial degree.
  SK-POLY-DEGREE-INFLECTION
- I — Use turning points to establish a minimum possible polynomial degree.
  SK-POLY-DEGREE-TURNING
- I — Use zeros and their multiplicities to establish a minimum possible polynomial degree.
  SK-POLY-DEGREE-ZEROS
- A — Infer polynomial end behavior from degree and leading coefficient.
  SK-POLY-END-BEHAVIOR · first introduced in Math 39

M49-POL-001

I can build or select a polynomial model for a context and interpret its features.

Textbook: 3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

- D — Select a function family whose behavior fits a table, graph, or context.
  SK-MODEL-FAMILY-SELECT · first introduced in Math 39
- D — Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A — Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A — Build a polynomial from zeros, multiplicities, and a point.
  SK-POLY-CONSTRUCT · first introduced in Math 31

Rational Functions and Limits

REQUIRED

Textbook coverage: 3.7 Rational Functions; 12.1 Finding Limits: Numerical and Graphical Approaches; 2.1 A Preview of Calculus; 2.2 The Limit of a Function; 2.3 The Limit Laws; 3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions; 4.4 Graphs of Logarithmic Functions; Chapter 2 Limits; 3.9 Modeling Using Variation

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M49-ALG-006

I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.

Textbook: 3.7 Rational Functions

SUPPORTING SKILLS

- A Cancel common factors rather than individual terms.
  SK-RAT-CANCEL-FACTORS · first introduced in Math 21
- I Determine horizontal or slant asymptotic behavior.
  SK-RAT-END-BEHAVIOR
- A Factor numerators and denominators completely.
  SK-RAT-FACTOR · first introduced in Math 21
- I Recall and correctly use rational-function terms including restriction, hole, asymptote, and end behavior.
  SK-RAT-FUNCTION-VOCAB
- I Identify a removable discontinuity and its coordinates.
  SK-RAT-HOLE
- I Determine vertical asymptotes from noncanceling denominator zeros.
  SK-RAT-VERTICAL-ASYMPTOTE
- R Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.
  SK-RAT-VOCABULARY · first introduced in Math 21

M49-ALG-007

I can sketch and analyze a rational function using its algebraic and asymptotic features.

Textbook: 3.7 Rational Functions

SUPPORTING SKILLS

- D Determine horizontal or slant asymptotic behavior.
  SK-RAT-END-BEHAVIOR · first introduced in Math 49
- D Identify a removable discontinuity and its coordinates.
  SK-RAT-HOLE · first introduced in Math 49
- A Determine excluded values from an original denominator.
  SK-RAT-RESTRICTIONS · first introduced in Math 21
- I Determine the sign of a rational function on intervals.
  SK-RAT-SIGN-INTERVAL
- D Determine vertical asymptotes from noncanceling denominator zeros.
  SK-RAT-VERTICAL-ASYMPTOTE · first introduced in Math 49

M49-LIM-001

I can interpret limit notation as a statement about function behavior near an input.

Textbook: 12.1 Finding Limits: Numerical and Graphical Approaches; 2.1 A Preview of Calculus; 2.2 The Limit of a Function

SUPPORTING SKILLS

- I Read and write limit notation with correct approach direction.
  SK-LIM-NOTATION

M49-LIM-002

I can estimate a limit from a graph or table.

Textbook: 12.1 Finding Limits: Numerical and Graphical Approaches; 2.1 A Preview of Calculus; 2.2 The Limit of a Function

SUPPORTING SKILLS

- I — Estimate one-sided and two-sided limits from a graph.
  SK-LIM-GRAPH
- I — Estimate a limit from input-output values approaching from both sides.
  SK-LIM-TABLE

M49-LIM-003

I can evaluate a limit algebraically using direct substitution or simplification.

Textbook: 2.1 A Preview of Calculus; 2.3 The Limit Laws

SUPPORTING SKILLS

- I — Resolve an indeterminate algebraic form by factoring or rationalizing.
  SK-LIM-SIMPLIFY
- I — Test a limit using direct substitution.
  SK-LIM-SUBSTITUTE
- A — Factor numerators and denominators completely.
  SK-RAT-FACTOR · first introduced in Math 21

M49-LIM-005

I can interpret an infinite limit as vertical-asymptotic behavior.

Textbook: 2.1 A Preview of Calculus; 2.2 The Limit of a Function

SUPPORTING SKILLS

- I — Interpret unbounded output near a finite input as an infinite limit.
  SK-LIM-INFINITE
- D — Determine vertical asymptotes from noncanceling denominator zeros.
  SK-RAT-VERTICAL-ASYMPTOTE · first introduced in Math 49

M49-LIM-006

I can use limit notation to describe behavior at infinity.

Textbook: 3.7 Rational Functions; 2.1 A Preview of Calculus; 2.2 The Limit of a Function

SUPPORTING SKILLS

- I — Compare dominant terms to determine behavior at infinity.
  SK-LIM-INFINITY
- D — Determine horizontal or slant asymptotic behavior.
  SK-RAT-END-BEHAVIOR · first introduced in Math 49

M49-LIM-007

I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions, and combinations thereof.

Textbook: 3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions; 3.7 Rational Functions; 4.4 Graphs of Logarithmic Functions; 2.1 A Preview of Calculus; Chapter 2 Limits

SUPPORTING SKILLS

- A Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP · first introduced in Math 21
- D Compare the long-run growth or decay of functions from different families.
  SK-FUNC-DOMINANCE · first introduced in Math 39
- I Apply the long-run growth hierarchy among logarithmic, polynomial, and exponential functions.
  SK-LIM-GROWTH-ORDER
- A Compare dominant terms to determine behavior at infinity.
  SK-LIM-INFINITY · first introduced in Math 49

M49-ALG-008

I can build or select a rational or rational-adjacent model for a context and interpret its long-run behavior.

Textbook: 3.7 Rational Functions; 3.9 Modeling Using Variation

SUPPORTING SKILLS

- D Select a function family whose behavior fits a table, graph, or context.
  SK-MODEL-FAMILY-SELECT · first introduced in Math 39
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A Determine horizontal or slant asymptotic behavior.
  SK-RAT-END-BEHAVIOR · first introduced in Math 49

Sequences and Series REQUIRED

Textbook coverage: 11.1 Sequences and Their Notations; 11.2 Arithmetic Sequences; 11.3 Geometric Sequences; 11.4 Series and Their Notations

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M49-SEQ-001

I can evaluate and graph a sequence defined explicitly or recursively.

Textbook: 11.1 Sequences and Their Notations

SUPPORTING SKILLS

- I — Generate an explicit formula from a sequence pattern.
  SK-SEQ-EXPLICIT
- I — Graph a sequence as discrete ordered pairs.
  SK-SEQ-GRAPH
- I — Interpret subscripts and evaluate terms of a sequence.
  SK-SEQ-NOTATION
- I — Generate a recursive formula with an initial condition.
  SK-SEQ-RECURSIVE
- I — Recall and correctly use the terms sequence, term, index, explicit formula, recursive formula, arithmetic sequence, and geometric sequence.
  SK-SEQ-VOCABULARY

M49-SEQ-002

I can write explicit and recursive formulas for arithmetic sequences.

Textbook: 11.2 Arithmetic Sequences

SUPPORTING SKILLS

- I — Recognize constant difference in an arithmetic sequence.
  SK-SEQ-DIFFERENCE
- D — Generate an explicit formula from a sequence pattern.
  SK-SEQ-EXPLICIT · first introduced in Math 49
- D — Generate a recursive formula with an initial condition.
  SK-SEQ-RECURSIVE · first introduced in Math 49

M49-SEQ-003

I can write explicit and recursive formulas for geometric sequences.

Textbook: 11.3 Geometric Sequences

SUPPORTING SKILLS

- D — Generate an explicit formula from a sequence pattern.
  SK-SEQ-EXPLICIT · first introduced in Math 49
- I — Recognize constant ratio in a geometric sequence.
  SK-SEQ-RATIO
- D — Generate a recursive formula with an initial condition.
  SK-SEQ-RECURSIVE · first introduced in Math 49

M49-SEQ-004

I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.

Textbook: 11.1 Sequences and Their Notations; 11.2 Arithmetic Sequences; 11.3 Geometric Sequences

SUPPORTING SKILLS

- A Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP · first introduced in Math 21
- D Recognize constant difference in an arithmetic sequence.
  SK-SEQ-DIFFERENCE · first introduced in Math 49
- D Recognize constant ratio in a geometric sequence.
  SK-SEQ-RATIO · first introduced in Math 49

M49-SEQ-005

I can create and interpret a sequence model for discrete change.

Textbook: 11.2 Arithmetic Sequences; 11.3 Geometric Sequences

SUPPORTING SKILLS

- A Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Generate an explicit formula from a sequence pattern.
  SK-SEQ-EXPLICIT · first introduced in Math 49
- I Identify initial value and pattern of change in a discrete context.
  SK-SEQ-MODEL
- A Generate a recursive formula with an initial condition.
  SK-SEQ-RECURSIVE · first introduced in Math 49

M49-SER-001

I can interpret and use summation notation.

Textbook: 11.4 Series and Their Notations

SUPPORTING SKILLS

- A Interpret subscripts and evaluate terms of a sequence.
  SK-SEQ-NOTATION · first introduced in Math 49
- I Expand and evaluate a finite sum written in sigma notation.
  SK-SER-SIGMA
- I Recall and correctly use the terms series, partial sum, sigma notation, finite series, infinite series, and convergence.
  SK-SER-VOCABULARY

M49-SER-002

I can calculate a finite arithmetic series.

Textbook: 11.4 Series and Their Notations

SUPPORTING SKILLS

- A Recognize constant difference in an arithmetic sequence.
  SK-SEQ-DIFFERENCE · first introduced in Math 49
- I Apply an arithmetic-series sum formula.
  SK-SER-ARITHMETIC

M49-SER-003

I can calculate a finite geometric series.

Textbook: 11.4 Series and Their Notations

SUPPORTING SKILLS

- A Recognize constant ratio in a geometric sequence.
  SK-SEQ-RATIO · first introduced in Math 49
- I Apply a finite geometric-series sum formula.
  SK-SER-GEOMETRIC-FINITE

M49-SER-004

I can determine whether an infinite geometric series converges and find its sum when it does.

Textbook: 11.4 Series and Their Notations

SUPPORTING SKILLS

- A Recognize constant ratio in a geometric sequence.
  SK-SEQ-RATIO · first introduced in Math 49
- I Test convergence and sum an infinite geometric series.
  SK-SER-GEOMETRIC-INFINITE

M49-SER-005

I can create and interpret a series model for accumulated change.

Textbook: 11.4 Series and Their Notations

SUPPORTING SKILLS

- A Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Apply an arithmetic-series sum formula.
  SK-SER-ARITHMETIC · first introduced in Math 49
- A Apply a finite geometric-series sum formula.
  SK-SER-GEOMETRIC-FINITE · first introduced in Math 49
- A Test convergence and sum an infinite geometric series.
  SK-SER-GEOMETRIC-INFINITE · first introduced in Math 49
- I Identify the initial term, change factor, and number of terms in an application.
  SK-SER-MODEL

Additional Limit Concepts EXTENSION

Textbook coverage: 12.1 Finding Limits: Numerical and Graphical Approaches; 2.1 A Preview of Calculus; 2.2 The Limit of a Function; 2.4 Continuity

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M49-LIM-004

I can determine one-sided limits and decide whether a two-sided limit exists.

Textbook: 12.1 Finding Limits: Numerical and Graphical Approaches; 2.1 A Preview of Calculus; 2.2 The Limit of a Function

SUPPORTING SKILLS

- D Estimate one-sided and two-sided limits from a graph.
  SK-LIM-GRAPH · first introduced in Math 49
- D Read and write limit notation with correct approach direction.
  SK-LIM-NOTATION · first introduced in Math 49
- A Estimate a limit from input-output values approaching from both sides.
  SK-LIM-TABLE · first introduced in Math 49

M49-LIM-008

I can distinguish a function value from a limit and determine continuity at a point.

Textbook: 12.1 Finding Limits: Numerical and Graphical Approaches; 2.1 A Preview of Calculus; 2.4 Continuity

SUPPORTING SKILLS

- I Check the three conditions for continuity at a point.
  SK-LIM-CONTINUITY
- A Estimate one-sided and two-sided limits from a graph.
  SK-LIM-GRAPH · first introduced in Math 49
- D Test a limit using direct substitution.
  SK-LIM-SUBSTITUTE · first introduced in Math 49

Parametric Equations

EXTENSION

Textbook coverage: 8.7 Parametric Equations: Graphs; 8.6 Parametric Equations

M49-PAR-001

I can graph a plane curve defined by parametric equations with its orientation.

Textbook: 8.7 Parametric Equations: Graphs

SUPPORTING SKILLS

- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Graph a parametric curve with its direction of motion.
  SK-PAR-GRAPH · first introduced in Math 32
- D Generate ordered pairs from parametric equations.
  SK-PAR-TABLE · first introduced in Math 32

M49-PAR-002

I can eliminate a parameter to obtain a rectangular equation.

Textbook: 8.6 Parametric Equations

SUPPORTING SKILLS

- A Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- D Eliminate a parameter algebraically.
  SK-PAR-ELIMINATE · first introduced in Math 32
- A Substitute an equivalent expression for a variable.
  SK-SYS-SUBSTITUTE · first introduced in Math 12

M49-PAR-003

I can create or interpret parametric equations for motion in the plane.

Textbook: 8.6 Parametric Equations; 8.7 Parametric Equations: Graphs

SUPPORTING SKILLS

- D Interpret or construct parametric equations for planar motion.
  SK-PAR-MODEL · first introduced in Math 32
- A Generate ordered pairs from parametric equations.
  SK-PAR-TABLE · first introduced in Math 32
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

Conic Sections

EXTENSION

Textbook coverage: 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates; 10.3 The Parabola

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M49-CON-001

I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.

Textbook: 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

- I — Classify a circle, ellipse, hyperbola, or parabola from an equation or geometric definition.
  SK-CONIC-CLASSIFY
- I — Use fixed-distance and focal-distance relationships to explain circles, ellipses, and hyperbolas.
  SK-CONIC-DISTANCE-DEFINITION
- I — Recall the defining terms and standard notation for circles, parabolas, ellipses, and hyperbolas.
  SK-CONIC-VOCABULARY

M49-CON-002

I can use completing the square to rewrite a general-form conic equation in standard form.

Textbook: 10.1 The Ellipse; 10.2 The Hyperbola; 10.3 The Parabola

SUPPORTING SKILLS

- A — Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I — Complete squares to convert a general conic equation to standard form.
  SK-CONIC-GENERAL-STANDARD
- D — Create a perfect-square trinomial by completing the square.
  SK-QUAD-COMPLETE-SQUARE · first introduced in Math 21

M49-CON-003

I can write, interpret, and graph implicit and parametric equations of a circle.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- I — Connect implicit, standard, and parametric equations of a circle.
  SK-CONIC-CIRCLE-EQUATIONS
- I — Determine and use a circle's center and radius.
  SK-CONIC-CIRCLE-FEATURES
- A — Graph a parametric curve with its direction of motion.
  SK-PAR-GRAPH · first introduced in Math 32
- A — Generate ordered pairs from parametric equations.
  SK-PAR-TABLE · first introduced in Math 32

M49-CON-004

I can write, interpret, and graph implicit and parametric equations of an ellipse.

Textbook: 10.1 The Ellipse

SUPPORTING SKILLS

- I Connect implicit, standard, and parametric equations of an ellipse.
  SK-CONIC-ELLIPSE-EQUATIONS
- I Determine and use an ellipse's center, axes, vertices, and foci.
  SK-CONIC-ELLIPSE-FEATURES
- A Graph a parametric curve with its direction of motion.
  SK-PAR-GRAPH · first introduced in Math 32
- A Generate ordered pairs from parametric equations.
  SK-PAR-TABLE · first introduced in Math 32

M49-CON-005

I can write, interpret, and graph implicit and parametric equations of a hyperbola.

Textbook: 10.2 The Hyperbola

SUPPORTING SKILLS

- I Connect implicit, standard, and parametric equations of a hyperbola.
  SK-CONIC-HYPERBOLA-EQUATIONS
- I Determine and use a hyperbola's center, axes, vertices, foci, and asymptotes.
  SK-CONIC-HYPERBOLA-FEATURES
- A Graph a parametric curve with its direction of motion.
  SK-PAR-GRAPH · first introduced in Math 32
- A Generate ordered pairs from parametric equations.
  SK-PAR-TABLE · first introduced in Math 32

M49-CON-006

I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.

Textbook: 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

- D Determine and use a circle's center and radius.
  SK-CONIC-CIRCLE-FEATURES · first introduced in Math 49
- A Classify a circle, ellipse, hyperbola, or parabola from an equation or geometric definition.
  SK-CONIC-CLASSIFY · first introduced in Math 49
- D Determine and use an ellipse's center, axes, vertices, and foci.
  SK-CONIC-ELLIPSE-FEATURES · first introduced in Math 49
- D Determine and use a hyperbola's center, axes, vertices, foci, and asymptotes.
  SK-CONIC-HYPERBOLA-FEATURES · first introduced in Math 49

M49-CON-007

I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.

Textbook: 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

- D Use fixed-distance and focal-distance relationships to explain circles, ellipses, and hyperbolas.
  SK-CONIC-DISTANCE-DEFINITION · first introduced in Math 49
- A Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.
  SK-COORD-DISTANCE-PYTHAGOREAN · first introduced in Math 22

M49-CON-008

I can explain a parabola as the set of points equidistant from a focus and directrix.

Textbook: 10.3 The Parabola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

- I Explain a parabola using its focus-directrix distance relationship.
  SK-CONIC-PARABOLA-DEFINITION
- A Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.
  SK-COORD-DISTANCE-PYTHAGOREAN · first introduced in Math 22